ECON 0100 | Exercise E2 | Competitive Firms

Exercises are a certificate of the work done together in class and offers extra practice outside class.

Q1 | Choosing Quantity

The animation's competitive firm has cost functions: $TC = 1000 + q^2$ and $MC = 2q$.

a) If the market price is $P = 80$, find:

- Total revenue from selling 20 units: TR_{20}
- Total revenue selling 21 units: TR_{21}

What is marginal revenue?

b) At $P = 80$, what quantity maximizes the firm's profit? Why not 30 or 50 units?

c) Calculate the firm's maximal profit. (*Hint: ATC at the profit-maximizing quantity is 65.*)

Q2 | Operating or Leaving

At a price of 40, the best positive output is 20 units. Fixed cost is unavoidable this period but avoidable in future periods by leaving the industry.

a) Calculate profit from operating and from shutting down this period. Which should the firm choose? (*Hint: ATC at the profit-maximizing quantity is 70.*)

b) If this price persists, should the firm stay in the industry in the long run? Explain why this answer can differ from (a).

c) At a price of 40, would firms enter or exit? What happens to market supply and price?

Q3 | A Higher Marginal Cost

Now suppose every firm's marginal costs go up. Start with identical firms earning zero economic profit in long-run equilibrium. Every unit now costs the same additional amount to produce. Fixed cost and market demand are unchanged, demand slopes down, and entry and exit are free.

a) What happens to price, each firm's output, and profit after the new cost?

b) What happens in the long run to price, each remaining firm's output, total market output, and the number of firms?

The Market

One Firm